

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular/Supplementary Winter Examination – 2024 Course: B.Tech. Branch : Electrical Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Semester : VII Subject Code & Name: BTEEE705C: ELECTRICAL POWER QUALITY Max Marks: 60 Date: 21/02/2025 Duration: 3 Hrs				
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 7. 4. The level of question/expected answer as per CPE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.				
			(CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)			12
1	What is the primary purpose of power quality evaluation?			CO1 1
	a) To calculate energy losses	b) To identify and address potential issues in voltage and current delivery	c) To measure the efficiency of power generation	d) To design electrical systems for maximum output
	Which of the following is NOT a type of power quality problem?			CO1 1
	a) Voltage fluctuation	b) Current fluctuation	c) Voltage imbalance	d) Waveform distortion
3	What does CBEMA in the CBEMA curve stand for?			CO1 1
	a) Computer Business Equipment Manufacturers Association	b) Consumer Bureau of Electrical Maintenance Association	c) Current Based Electrical Maintenance Agency	d) Circuit Breaker Equipment and Machinery Association
4	What is one key measure to manage ferro resonance?			CO2 1
	a) Avoid lightly loaded transformers	b) Use larger transformers	c) Increase harmonic levels	d) Shorten transmission lines
	Which phenomenon is a common source of ferro resonance?			CO2 1
	a) Saturated transformer cores	b) High-voltage lightning strikes	c) Sudden power outages	d) Capacitor switching
6	Which principle is critical for controlling harmonics?			CO3 1
	a) Increasing transformer size	b) Isolating nonlinear loads	c) Using filters to suppress harmonic currents	d) Reducing transmission line length
7	What is the relationship between nonlinear loads and harmonic distortion?			CO4 1

	a) Nonlinear loads are not affected by harmonic distortion	b) Nonlinear loads are the primary cause of harmonic distortion	c) Nonlinear loads suppress harmonic distortion	d) Nonlinear loads generate harmonics only during switching		
8	Why are distributed resources important for modern power systems?				CO4	1
	a) They reduce the need for centralized generation	b) They support local voltage regulation and improve system reliability	c) They minimize transmission line losses	d) They eliminate harmonic distortions		
9	How do utility voltage regulators maintain voltage stability?				CO4	1
	a) By reducing system frequency	b) By isolating nonlinear loads	c) By increasing the harmonic levels	d) By adjusting the transformer tap positions		
10	Which parameter is NOT typically measured in power quality monitoring?				CO5	1
	a) Voltage sags and swells	b) Harmonic distortion	c) Power factor	d) Line impedance		
11	What is the purpose of power quality monitoring standards?				CO5	1
	a) To specify the cost of monitoring equipment	b) To ensure uniformity and reliability in power quality data assessment	c) To promote energy conservation	d) To define the size of power quality analyzers		
12	What is the typical location of voltage regulators in a utility network?				CO3	1
	a) At the generation station	b) Along transmission and distribution lines	c) At end-user facilities	d) Near harmonic filters		
Q. 2	Solve the following.					12
A)	Explain power quality evaluation procedure.				CO1	6
B)	Briefly describe about (i) Impulsive transients (ii) Oscillatory transients				CO1	6

Q.3	Solve the following.		12
A)	What are fundamental principles of over voltage protection of load equipment? Explain them in brief?	CO2	6
B)	Explain briefly about long duration and short duration voltage variations.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	Define flicker and explain sources of flicker.		6
B)	Describe the various devices used for voltage regulation and their applications.		6
C)	Discuss the role of capacitors in voltage regulation for both utility applications and end-users		6
Q.5	Solve Any Two of the following.		12
A)	What are common wiring and grounding problems encountered in electrical installations? Provide examples and their potential impacts.	CO4	6
B)	Describe briefly Voltage distortion and Current distortion.	CO4	6
C)	Explain various power quality monitoring standards.	CO4	6
Q.6	Solve Any Two of the following.		12
A)	Explain the primary reasons for grounding in electrical systems. How does grounding enhance safety and system reliability?	CO5	6
B)	Evaluate the importance of intelligent systems in assessing power quality measurement data. How do they enhance monitoring effectiveness?	CO5	6
C)	List various types of power quality measuring equipment and explain any three types of power quality measuring equipment.	CO5	6
*** End ***			